

BATHURST, NB, Canada

WMO: 715740

Lat: 47.631N

Lon: 65.741W

Elev: 193

StdP: 14.59

Time Zone: -4.00 (NAA)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -11.5	(c) -6.2	(d) -18.7	(e) 2.0	(f) -9.6	(g) -14.3	(h) 2.5	(i) -4.5	(j) 25.7	(k) 18.6	(l) 23.1	(m) 18.4	(n) 4.8	(o) 260	(p) 0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	85.0	69.0	81.4	67.4	78.2	65.5	71.6	80.1	69.8	78.0	68.1	75.0	10.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.9	107.0	75.0	67.2	100.7	73.1	65.5	94.8	71.8	35.5	79.9	34.0	77.9	32.6	75.2	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.0	18.6	16.5	-21.4	90.9	4.8	2.4	-24.8	92.6	-27.7	94.0	-30.4	95.4	-33.9	97.1		
			-21.6	75.0	4.8	1.5	-25.1	76.1	-27.9	77.0	-30.6	77.8	-34.1	78.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	14.4	15.3	24.6	37.3	49.7	58.9	66.9	65.0	56.6	45.4	33.4	21.3	
	DBStd	20.04	11.72	10.95	9.96	7.05	7.08	6.25	5.44	5.07	6.80	7.16	8.85	10.15	
	HDD50	4934	1104	972	788	387	93	4	0	0	16	180	501	889	
	HDD65	9001	1569	1392	1252	833	477	203	41	63	261	609	947	1354	
	CDD50	1606	0	0	1	4	84	271	524	466	215	37	4	0	
	CDD65	198	0	0	0	0	3	20	100	64	10	1	0	0	
	CDH74	1826	0	0	0	4	73	248	861	489	144	7	0	0	
	CDH80	440	0	0	0	0	14	59	229	104	33	1	0	0	
Wind	WSAvg	6.6	7.6	7.8	8.0	7.1	6.4	5.6	5.3	5.2	5.8	6.7	7.0	7.4	
Precipitation	PrecAvg	42.8	3.6	2.7	3.3	3.6	3.2	3.2	3.8	4.0	2.9	3.6	4.2	4.4	
	PrecMax	55.7	6.3	4.6	6.5	6.8	6.3	5.8	9.4	8.6	6.1	8.5	8.0	6.8	
	PrecMin	33.7	0.1	1.0	1.0	0.9	1.0	1.6	0.6	1.3	1.0	0.9	1.8	1.7	
	PrecStd	5.9	1.7	1.1	1.4	1.8	1.6	1.3	1.9	1.9	1.3	2.0	1.6	1.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.1	43.9	59.3	70.2	81.3	86.2	89.9	87.4	84.8	73.0	63.6	48.5	
		MCWB	46.2	40.5	50.4	54.6	63.6	67.5	71.9	69.8	69.8	61.4	60.6	46.3	
	2%	DB	39.2	37.7	46.0	59.4	74.7	80.7	85.7	83.0	77.9	66.4	56.3	42.1	
		MCWB	37.1	34.2	39.5	48.4	59.3	65.2	69.6	69.0	65.5	59.0	52.0	40.0	
	5%	DB	35.5	34.2	41.7	53.7	68.5	76.1	82.2	79.2	73.4	61.5	50.6	38.3	
		MCWB	33.7	32.1	36.2	44.8	55.4	62.3	68.6	66.8	63.4	55.1	46.4	36.3	
	10%	DB	31.7	31.6	38.2	49.2	63.6	71.9	78.6	76.0	69.7	57.3	46.1	34.8	
		MCWB	30.3	29.9	33.6	41.6	53.3	60.5	66.2	65.3	61.7	51.9	42.9	33.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	46.2	41.2	50.5	57.4	66.5	70.9	74.5	72.9	71.6	64.5	61.3	47.1
			MCDB	47.3	43.9	56.4	67.1	77.6	80.8	83.4	81.0	78.7	67.7	63.3	48.2
2%		WB	37.6	35.3	40.4	49.9	61.7	67.8	72.3	70.9	68.0	60.6	53.6	40.7	
		MCDB	38.7	36.5	44.5	56.5	70.4	76.5	80.9	79.7	73.5	65.5	55.6	41.4	
5%		WB	33.7	32.3	36.8	45.8	58.1	65.1	70.3	69.0	65.8	56.3	46.8	36.7	
		MCDB	35.5	33.8	40.9	52.6	65.5	72.7	78.3	76.5	71.9	60.3	48.7	38.1	
10%		WB	30.4	29.8	33.9	42.4	54.5	62.5	68.8	67.3	62.8	52.7	42.9	33.5	
		MCDB	31.7	31.4	37.5	48.1	62.2	69.3	76.1	73.4	68.6	56.1	45.8	34.5	
Mean Daily Temperature Range	5% DB	MDBR	17.6	18.4	17.6	16.7	20.4	19.3	19.8	19.3	20.2	16.3	13.6	13.9	
		MCDBR	18.6	18.6	21.7	27.6	30.1	28.4	25.7	24.7	26.3	23.9	20.1	15.4	
		MCWBR	17.5	16.2	16.4	17.6	17.2	15.1	12.9	12.9	16.2	17.0	16.9	14.5	
	5% WB	MCDBR	18.2	16.9	19.4	24.5	26.3	22.8	20.8	21.2	22.7	20.8	18.2	15.0	
		MCWBR	17.3	15.3	15.6	17.3	17.0	13.4	11.7	12.0	15.7	16.5	16.9	14.4	
Clear-Sky Solar Irradiance	taub	0.277	0.278	0.314	0.379	0.366	0.386	0.418	0.391	0.361	0.331	0.297	0.279		
	taud	2.355	2.325	2.331	2.259	2.387	2.344	2.288	2.358	2.432	2.507	2.532	2.435		
	Ebn at Noon	256	283	288	276	285	279	267	270	269	261	250	243		
	Edh at Noon	22	28	33	40	36	38	40	36	30	24	19	18		
All-Sky Solar Radiation	RadAvg	406	679	1064	1377	1623	1753	1740	1522	1178	727	395	293		
	RadStd	48	77	78	117	149	133	135	95	98	61	50	32		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page